

PCA

Mithilesh singh

What is Principal Component Analysis?

Principal Component Analysis technique used for EDA exploratory data analysis, especially in dimensionality reduction.

How does Principal Component Analysis work?

PCA achieves dimensionality reduction by **finding the principal components** of the data points.

The principal components are the **vectors of highest variance**.

In other words, they are the directions where the data points vary the most. PCA finds these vectors that explain most of the variance observed in the data points.

PCA Intuition

If you have a large data sets with thousands/millions of observations (rows) and hundreds of different variables (columns)

While having more data is always great, sometimes they have so much information in them, we would have impossibly long model training time and the curse of dimensionality starts to become a problem.

Direct elimination of variables is the obvious way but it has clearly impact on the *information* content of your data set. Too much or a wrong eliminations and your data set becomes useless, too less and the data set remains large and difficult to analyze.

This is our **first assumption** to keep in mind: ***higher variance => higher information content***. Only the context and our data can tell if this assumption holds.

Variance is Both Our Enemy and Our Friend

We can compare PCA with writing a book summary.

Finding the time to read a 1000-pages book is a luxury that few can afford. Wouldn't it be nice if we can summarize the most important points in just 2 or 3 pages so that the information is easily digestible?

We may lose some information in the process, but, at least we get the big picture.

How does PCA work?

It's a two-step process. We can't write a book summary if we haven't read or understood the content of the book.

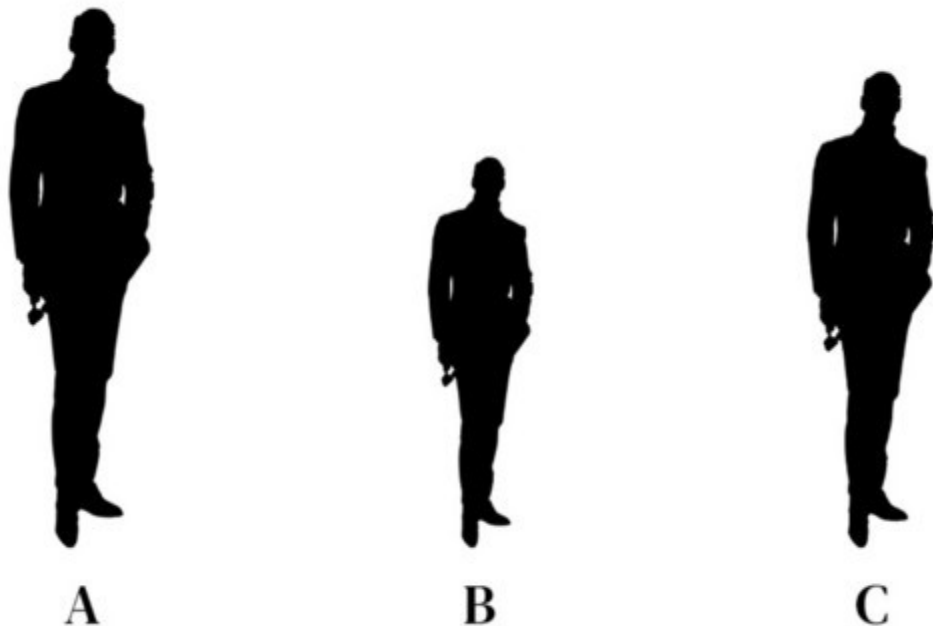
PCA works the same way — understand, then summarize.

Understanding data, the PCA way

- . Can PCA understand which part of our data is important?
- . Can we mathematically quantify the amount of information embedded within the data?

yes, *variance* can.

The greater the variance, the more the information. Vice versa.



The silhouette of three identical friends whom we need to identify based on their height differences.

Without a doubt, we are going to say that Person A, Person B, and Person C.

Now, let's try and guess a different group of friends.

Another group of our friends' heights that we remember by heart.

.



A



B



C

The silhouette of three similarly-tall friends whom we need to identify.

In the same way, when our data has a higher variance, it holds more information. This is why we keep hearing PCA and maximum variance in the same sentence.

*PCA is defined as an orthogonal linear transformation that transforms the data to a new coordinate system such that the **greatest variance** by some scalar projection of the data comes to lie on the first coordinate (called the first principal component), the second greatest variance on the second coordinate, and so on.*

In the eyes of PCA, variance is an objective and mathematical way to quantify the amount of information in our data.

Variance is information.

In addition to focusing on the high variance data concept a **second assumption: correlation between variables is a form of data redundancy**. If there's a clear defined relation between two variables like for example height of a person , weight of a person then one of the two variables are useless and can be eliminated directly from the data set. When instead the dependencies are less clear direct elimination is not recommended and one can try to manage the lack of evidence for correlation through a third assumption.

The **third assumption** is: ***we assume that whatever is the correlation between the variables, this is linear.***

So to recap we are assuming that:

- variance is related to information content and should be maximized

- redundant variables and high correlations between variables are a form of noise that should be minimized
- correlations between variables are linear

guess who's who based on their height and weight.

Round two.

Person	Height (cm)	Weight (kg)
Alex	145	68
Ben	160	67
Chris	185	69

The same set of friends and their respective height and weight.

In the beginning, we only had height. Now, we have more amount of data(weight and height).

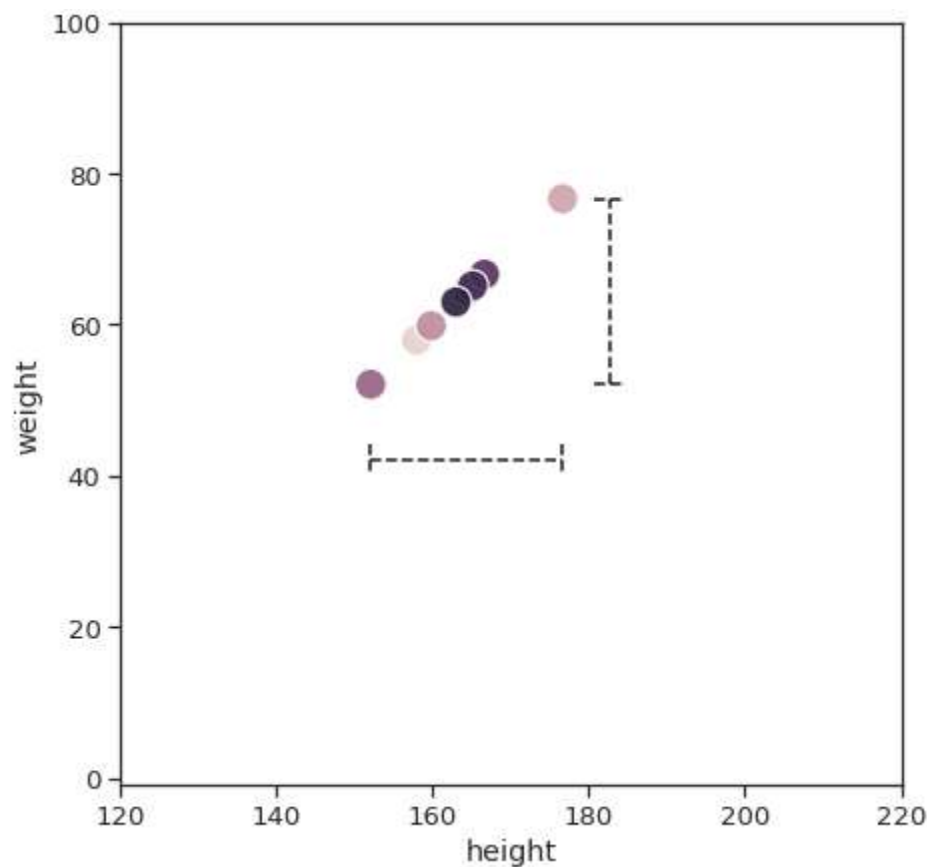
— how PCA summarizes our data, or more accurately, reduces dimensionality.

Summarizing data with PCA

Personally, the weight differences are so small (a.k.a small variance), it doesn't help me differentiate our friends at all. I still had to rely mostly on height to make my guesses.

Intuitively, we have just reduced our data from 2-dimensions to 1-dimension. The idea is that we can selectively keep the variables with higher variances and then forget about the variables with lower variance.

But what if-, just what if height and weight have the *same* variance? Does it mean we can no longer reduce the dimensionality of this data set?



The dotted line represents the variance of height and weight.

Feature	Variance
Height	1.11
Weight	1.11
TOTAL	2.22

All the features are standardized to the same scale for a fair comparison. Image by the author.

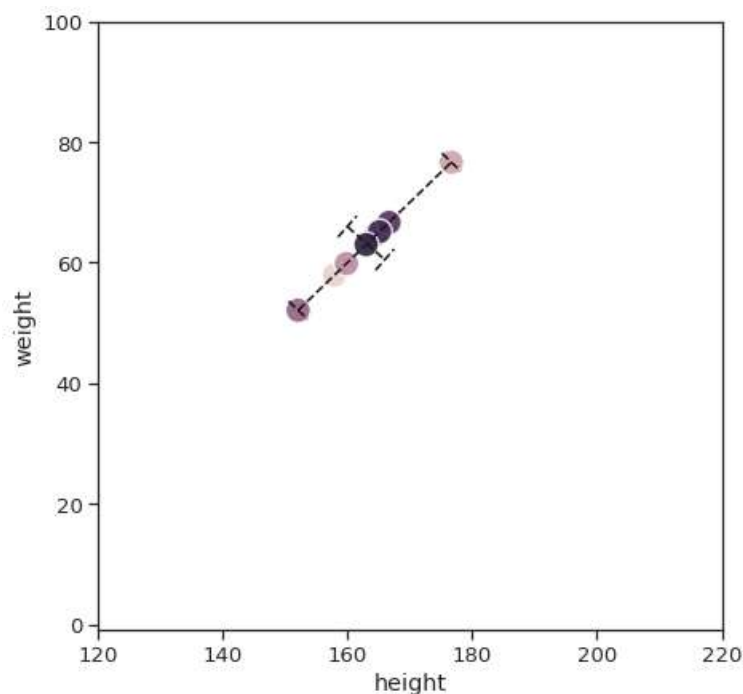
In this case, it's very difficult to choose the variables we want to delete. If I throw away either one of the variables, we are throwing away half of the information.

Can we keep *both*?

Perhaps, with a different perspective.

Up until now, we have only been looking at the variance of height and weight individually. Instead of limiting ourselves to choose just one or the other, why not combine them?

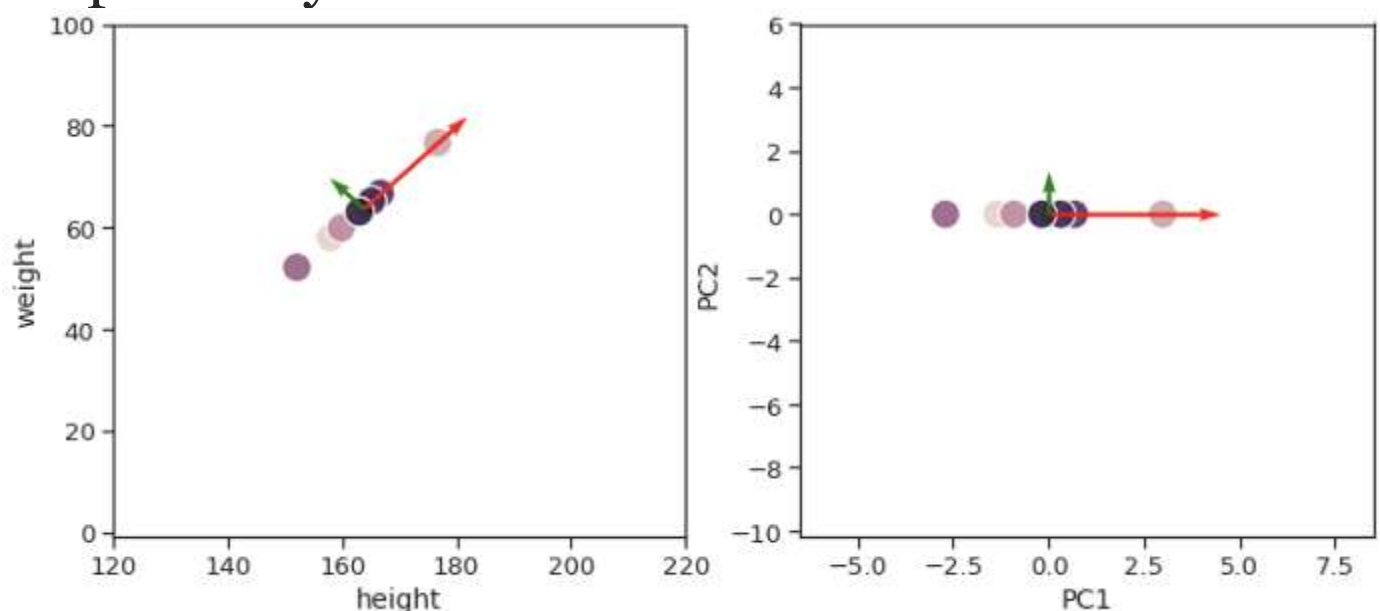
When we look closer at our data, the maximum amount of variance lies not in the x-axis, not in the y-axis, but a diagonal line across. The second-largest variance would be a line 90 degrees that cuts through the first.



The dotted line shows the direction of maximum variance.

To represent these 2 lines, PCA combines both height and weight to create two brand new variables. It could be 30% height and 70% weight, or 87.2% height and 13.8% weight, or any other combinations depending on the data that we have.

These two new variables are called the **first principal component (PC1)** and the **second principal component (PC2)**. Rather than using height and weight on the two axes, we can use PC1 and PC2 respectively.



(Left) The red and green arrows are the principal axes in the original data.

(Right) The direction of the principal axes have been rotated to become the new x- and y-axis.

When it comes to real data, more often than not, we won't get a principal component that captures 100% of the variances.

Performing a PCA will give us N number of principal components, where N is equal to the dimensionality of our original data.

From this list of principal components, we generally choose the least number of principal components that would explain the most amount of our original data.

We know that for two variables covariance formula is (in case our dataset is a sample of data taken out of a population):

$$Cov(x, y) = \frac{1}{n} \sum_{i=1}^n [(x_i - \bar{x})(y_i - \bar{y})]$$

Covariance for not centered around the mean sample data (Source: Author)

Covariance represents a dispersion measure that include the concept of linear “**synchronicity**” between two variables in relation to their respective means.

That means measuring, for each point, how **the difference** between <the x coordinate of a point> and <the mean of all the x points>, is “**in synch**” with **the difference** between <the y coordinate for the same point> and <the mean of all the y points> and then averaging that number.

From Covariance to Covariance Matrix

Considering that the covariance formula above was just for 2 variables we can see the covariance matrix as the “big picture” of all the covariances between all of the variables in our data set.

$$C_x = \begin{vmatrix} \textcolor{red}{Var}(X_1) & Cov(X_1, X_2) & \dots & Cov(X_1, X_n) \\ Cov(X_2, X_1) & \textcolor{red}{Var}(X_2) & \dots & Cov(X_2, X_n) \\ \dots & \dots & \dots & \dots \\ Cov(X_n, X_1) & Cov(X_n, X_2) & \dots & \textcolor{red}{Var}(X_n) \end{vmatrix}$$

Covariance Matrix (also called in literature Variance/Covariance Matrix) (Source:Author)

When we calculate the Variance/Covariance Matrix of our original dataset (we call the original dataset X as a collection of many column vectors $X_1, X_2 \dots X_n$) we see the variances along the diagonal but we also see the joint covariance's in the off diagonal elements which are a (difficult to interpret) measure of the magnitude of the '**synchronicity**' between one variable and the others.

Thus, PCA is a method that brings together:

1. A measure of how each variable is associated with one another. (Covariance matrix.)
2. The directions in which our data are dispersed. (Eigenvectors.)
3. The relative importance of these different directions. (Eigenvalues.)

PCA combines our predictors and allows us to drop the Eigenvectors that are relatively unimportant.

Steps of PCA

The PCA algorithm consists of the following steps.

1. Standardizing data by subtracting the mean and dividing by the standard deviation
2. Calculate the Covariance matrix.
3. Calculate eigenvalues and eigenvectors
4. Merge the eigenvectors into a matrix and apply it to the data. This rotates and scales the data. The principal components are now aligned with the axes of our features.

5. Keep the new features which account for the most variation and discard the rest.

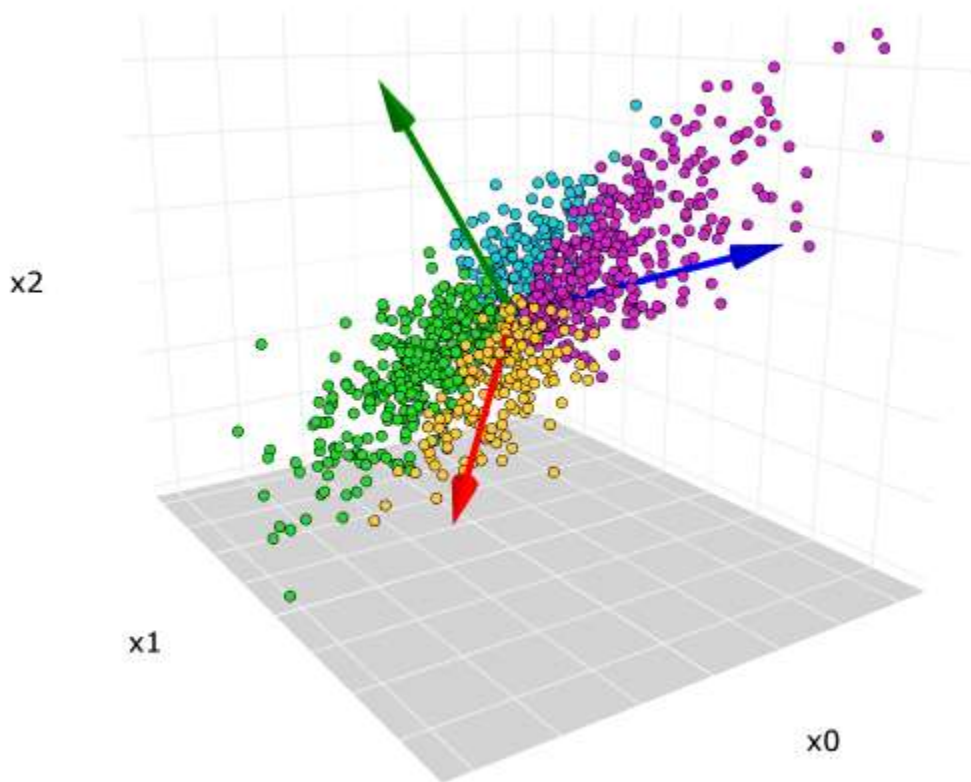
Applications of eigenvectors and eigenvalues values in Data Science

Principal components

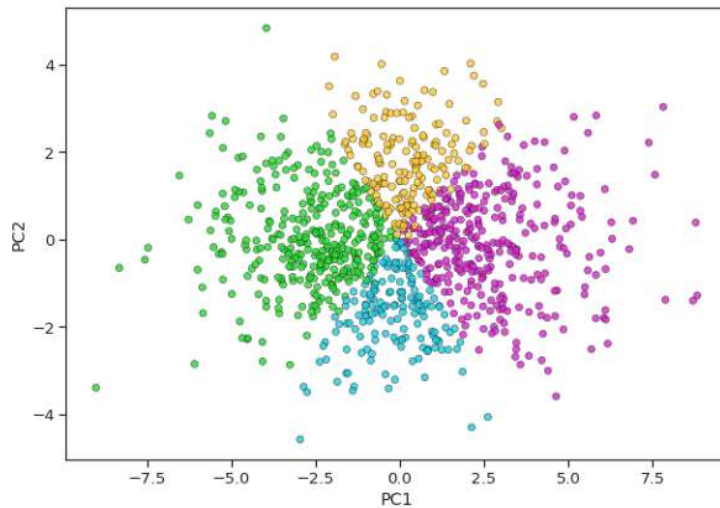
Using eigenvalues and eigenvectors, we can find the main axes of our data. The first main axis (also called “first principal component”) is the axis in which the data varies the most. The second main axis (also called “second principal component”) is the axis with the second largest variation and so on.

An another definition What is PCA?

Principal component analysis (PCA) is a technique that transforms high-dimensions data into lower-dimensions while retaining as much information as possible.



The original 3-dimensional data set. The red, blue, green arrows are the direction of the first, second, and third principal components, respectively.



Scatterplot after PCA reduced from 3-dimensions to 2-dimensions.

PCA is extremely useful when working with data sets that have a lot of features.